Supporting Information for: Pair Attribution of the VV10 Nonlocal-Correlation Term and Its Grid-Partition Dependence

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Provenance of every quoted number

Each tag used in the main text resolves to a file in this repository. "line N" means the value can be read literally at that line of the archived standard output; those stdout files live under `evidence/**`, are tracked, and are hash-listed under `evidence/*/MANIFEST.sha256`; they were not re-typed by hand. `nlcsplit/logs/` is a different case and is stated as such rather than glossed: only three sidecar JSONs there are tracked (`step_h.json`, `step_h_l2cp.json`, `orca_anchor.json`), the rest of that directory ignores `*.out` by its own `.gitignore`, so a run log left there is a working note and not evidence.

Table S1. Provenance appendix: what each [S#] tag resolves to, and which claim in the text it fixes.

Tag	Artefact	What it fixes
S1	`RESEARCH-20260929-lit.md` §1, §6	first-round WoS/Scopus search record: strings, hit counts, search date
S2	`MAIN-skeleton.md` tables 2 and 4b; `nlcsplit/logs/step_h.json`	the level-1 S22-subset table, incl. the 0.8% / 18.2% nonlocal fractions
S3	`nlcsplit/step_f_dispersion.py`	the level-1 dispersion-scale check that leaves the double sum as the only unaccounted term
S4	`MAIN-skeleton.md` table 4b	the 52%-97% range of the same quantity across the two archived geometries of one molecule pair (benzene dimer A parallel-displaced, B T-shaped), the geometry axis, not the partition axis
S5	`nlcsplit/step_e_kernel.py`	explicit pair sum vs PySCF's native VV10 kernel on a fixed grid
S6	`evidence/centos/20260928T165857Z_centos-step_c.out` line 19	relaxed O-O = 2.901 A
S7	`nlcsplit/step_d_floor.py`, `evidence/recompute/20260928T150222Z_step_d_floor.out`, `nlcsplit/tests/test_kernel.py`	the local term is beta x electron number, so it cancels in any superposition difference; the archived stdout is the rho_floor sweep itself - dropped-point fractions, electrons carried by the dropped points, and the per-level Ha spans behind the sensitivity claim in §3

Tag	Artefact	What it fixes
S8	`nlcsplit/logs/step_h.json`, `nlcsplit/logs/step_h_l2cp.json`, `evidence/l2cp/20260929T051912Z_centos-l2_benzmeth_cp.out`, `evidence/l2cp/20260929T051849Z_l2_benzmeth-wsl.out`, `evidence/centos/20260928T165857Z_centos-step_f_h2o2.out`, `evidence/recompute/20260929T010500Z_step1_geomfix.out`, `evidence/l2cp/20260929T051849Z_wsl2121-benzmeth-l2cp.json`, `evidence/l2cp/20260929T051912Z_centos2140-benzmeth-l2cp.json`, `evidence/l2cp/cross_version_pair.py`, `evidence/l2cp/20261004T0100Z_cross_version_pair_benzmeth.out`	cross-version pair (PySCF 2.12.1 vs 2.14.0) and the 19936 vs 20248 partitioned grid counts
S9	`nlcsplit/step_i_gridrepr.py`, `evidence/recompute/20260929T_step_i_gridrepr.out`	the two discrepancies that were previously reported as one number
S10	`evidence/l2cp/20260928T213839Z_centos-step_h_l2cp.out`, `evidence/l2cp/20260929T051912Z_centos-l2_benzmeth_cp.out`, `nlcsplit/logs/step_h_l2cp.json`	level-2 with counterpoise, all six S22 subsystems complete, including the `p_over_ds` column behind the 0.92-1.08 ratio
S11	`RESEARCH-20260929-lit.md` §6.2-6.3	published comparison value for JCTC 3c00903 (full text via ChemRxiv green OA 10.26434/chemrxiv-2023-nc64t)
S12	`evidence/recompute/20260930T191014Z_stepg-official.out` (official S22 geometry, the numbers quoted in sections 2 and 3.4), `evidence/recompute/20260929T003144Z_stepg-recompute.out` (constructed geometry, quoted as the cross-check), both produced by `nlcsplit/step_g_basis.py`, cross-checked against `evidence/centos/20260928T165857Z_centos-step2.out`	five-basis ladder of P_{frz} and of the four-scheme span at level 1, run once per geometry by the same script (`NCOFF` selects the official one). The two ladders agree to 5-8% on every span and to within 0.008 kcal/mol on the STO-3G-to-aug-cc-pVDZ drift, so the reading is geometry-insensitive; the script's own criterion (last-two-rung drift < 0.05) is satisfied on both geometries, which is what licenses calling the basis near-converged while the scheme span is not shrinking
S13	`evidence/centos/20260928T165857Z_centos-step2.out` lines 17-20, per-level values at lines 2-15	cross-grid spread per basis (sto-3g 0.0010, 6-31G* 0.0081, aug-cc-pVDZ 0.0067); the 0.0002 for 6-31G* over level 1 to 2 is the difference of lines 8 and 9
S14	Lazić, Atodiresei, Alaei, Caciuc, Blügel and Brako, *Comput. Phys. Commun.* **181**, 371-379 (2010), DOI 10.1016/j.cpc.2009.09.016; open preprint arXiv:0810.2273v1; judgement in `RESEARCH-20260929-lit.md` §10	the closest published decomposition of a nonlocal term, read in full text
S15	`evidence/retrieval_calibration/oa_stab.js`, `evidence/retrieval_calibration/openalex_stability.json`, `evidence/retrieval_calibration/xref_calib2.js`, `evidence/retrieval_calibration/xref_r3.js`	measured recall of the open indexes: 4/4 known titles at rank 0, 0/4 and 1/3 for legitimate topical strings
S16	`RESEARCH-20260929-lit.md` §8.1	the institutional-access precondition of the search, and why the first round was counted from the UI

Tag	Artefact	What it fixes
S17	`RESEARCH-20260929-lit.md` §13	second round: full enumeration through WoS's own record endpoint (RecordsSearched = 58,664,816, search date 2026-09-29), the three calibration strings reproduced record-for-record, the two zero-hit strings, complete readings of the two 4-record and four broad strings, and the three added citations whose DOIs were checked against both Crossref and WoS
S18	`nlcsplit/logs/orca_anchor.json`, `evidence/orca_run/h2o_dimer.out`, `evidence/orca_run/h2o_mono0.out`, `evidence/orca_run/h2o_mono1.out`, `evidence/orca_run/benz_dimer.out`, `evidence/orca_run/benz_mono0.out`, `evidence/orca_run/benz_mono1.out`, `evidence/orca_run/MANIFEST.sha256`, `evidence/orca_control/h2o_wb97x.out`, `evidence/orca_control/h2o_wb97xv.out`, `evidence/orca_control/MANIFEST.sha256`, `evidence/orca_widen/orca_widen.json`, `evidence/orca_widen/RECHECK_convergence.txt`, `evidence/orca_widen/generator_rescan_convergence.py`, `evidence/orca_widen/generator_merge_run.py`, `evidence/orca_widen/generator_cross_program_table.py`, `evidence/orca_widen/cross_program_table.txt`, `evidence/orca_widen/MANIFEST.sha256`	the cross-program referee: three ORCA 6.0.1 runs each (S22 official geometries, 6-31G*, every run with a converged SCF and exactly one `NL Energy, E(C,NL)` line), six for water and the benzene sandwich from `nlcsplit/logs/orca_anchor.json`, eighteen more archived under `evidence/orca_widen/` (one log per run, each listed with its sha256 in `evidence/orca_widen/MANIFEST.sha256`) covering ammonia, methane, benzene...water, benzene...ammonia, benzene...methane and the parallel-displaced benzene dimer; the per-file sha256 manifests; the ωB97X (no `-V`) control that prints no such line; the re-scan that corrected twelve falsely-flagged `converged` fields without touching any energy; and the generator that emits the eight-row comparison and refuses if the system list disagrees with the one it declares, or if any row's basis, grid level or counterpoise setting disagrees
S19	`evidence/centos/20260928T165857Z_centos-step_c.out` lines 6, 15 and 24	the level-0 ratios of the inter-fragment pair term to the superposition difference: 1.017 for the water dimer at the official S22 geometry, 1.025 at its ωB97X-V-optimised geometry, and 0.915 for the benzene sandwich - the third of these is a different system, which is why it is not folded into the 0.92-1.08 band
S20	`nlcsplit/step_k_population.py`, `evidence/population/20260929T131530Z_stepk_pop-official.out`, `evidence/population/20260929T131055Z_stepk_pop-constructed.out`	the level-3 four-scheme population table for the water dimer, from which beta*N_A is read; both runs assert that the fragment populations sum to the electron number. Re-running them is also what showed that the earlier tabulation of this number in the work log was labelled "official geometry" but reproduces only on the constructed one

Tag	Artefact	What it fixes
S21	`evidence/agentb_sweep/20260928T071951Z_agentb-sweep_631gstar.txt`, `evidence/agentb_sweep/20260929T151414Z_step_l_sweep_compare_levels0-9.out`, `evidence/agentb_sweep/20260929T151414Z_step_l_sweep_compare_levels0-9.json`, `nlcsplit/step_l_sweep_compare.py`	the independent sweep (water-monomer nonlocal term at levels 0-9, self-consistent and again on one fixed density matrix, ending in a partial `atom_grid` scan that dies on an unsupported angular grid), and the archived stdout plus sidecar JSON of our comparison against it, which carry the per-level difference column behind section 2.1 - the sweep file alone has no comparison column. Produced by the shipped script; its level-9 row covers the point pairs among 428,920 kept grid points, which is why that one row took hours. The `levels0-8` snapshot taken mid-run is kept beside it, hashed, as the record of what was known before level 9 landed; its header announces the intended 0-9 range, which is why that filename says 0-8 and this one says 0-9.

Tag	Artefact	What it fixes
S22	evidence/attribution_referee/run.py, evidence/attribution_referee/referee_lib.py, evidence/attribution_referee/summarize.py, evidence/attribution_referee/probe_quadrature.py, evidence/attribution_referee/README.md, evidence/attribution_referee/outputs/results.json, evidence/attribution_referee/outputs/table.md, evidence/attribution_referee/outputs/summary.md, evidence/attribution_referee/outputs/validation.md, evidence/attribution_referee/outputs/convergence.md, evidence/attribution_referee/outputs/headline.md, evidence/attribution_referee/outputs/run_log.txt, evidence/attribution_referee/outputs/results_run1_20261001.json, evidence/attribution_referee/outputs/table_run1_20261001.md, evidence/attribution_referee/outputs/rerun_verify_20261001.log, evidence/attribution_referee/MANIFEST.sha256, evidence/attribution_referee/hetero.py, evidence/attribution_referee/outputs/hetero.json, evidence/attribution_referee/outputs/hetero_run_20261001.log	the partition-free referee: two-Gaussian model density, the cross term by three independent routes (3-D Cartesian tensor product, 5-D cylindrical reduction, importance-sampled Monte-Carlo), the measured combined uncertainty 1.060e-05 Ha, the column that cancels the grid, and the falsification suite (positive control on the documented-wrong pairing, closed-form Gaussian and Coulomb anchors, node-normalisation guard). The two shipped runs give a file-level reproducibility check rather than a sentence: the two tables compare byte-identical, and the field-by-field diff of the two machine-readable results is confined to the recorded wall-clock fields and the worker count. The Monte-Carlo error bar in the same file comes from evidence/attribution_referee/mc_tighten.py (4,000,000 draws per configuration, its own seed base) and is stored in evidence/attribution_referee/outputs/mc_tight.json, a separate sidecar, so neither archived run is disturbed by it. Re-running evidence/attribution_referee/run.py regenerates every number; evidence/attribution_referee/summarize.py re-quotes them without re-quadrating. evidence/attribution_referee/hetero.py is the heteronuclear extension, and it is written as a sidecar for the same reason evidence/attribution_referee/mc_tighten.py is: it must not disturb the archived table. Row 0 is therefore homonuclear and labelled as the archived run labels it, and it reproduces the reference cross term and the package cross term with a difference of exactly 0.000e+00 - which is what licenses the three unequal-centre rows. It skips the referee's own disjoint half-box node set, because that construction assumes two congruent boxes, and it runs each density under two different element labels so the radii table's contribution is measured instead of assumed

Tag	Artefact	What it fixes
S23	`evidence/population/20260930T200927Z_stepk_soft-vs-hard-official.out`, `evidence/population/20260930T201142Z_stepk_soft-vs-hard-constructed.out`, `nlcsplit/step_k_population.py`, `evidence/population/MANIFEST.sha256`	the soft-vs-hard reading of the local term, per partition scheme, on both water-dimer geometries at grid level 3: hard populations credit a point to its owner only, soft populations spread it over the fragment factors at that point, and both are computed on PySCF's own partition weights so the measure is the converged one. The two files give the largest soft-hard population difference (1.527e-06 electrons official, 8.686e-07 constructed) and its conversion through beta into kcal/mol. Before these runs existed, this sentence quoted 8.7e-7 electrons on the strength of a working log only - the number was real but its geometry was not the published one, which is why the scope is now stated in the sentence

Tag	Artefact	What it fixes
S24	`nlcsplit/step_m_ccsdt.py`, `evidence/ccsdt/20260930T224400Z_stepm_ladder_after_refactor.out`, `evidence/ccsdt/20260930T235900Z_stepm_three_systems_631g.out`, `evidence/ccsdt/20261001T000400Z_stepm_three_systems_avdz.out`, `evidence/ccsdt/20260930T212300Z_stepm_ladder.out`, `evidence/ccsdt/20261001T044833Z_orca_crosscheck.out`, `evidence/ccsdt/20261001T052943Z_orca_crosscheck_three_systems.out`, `evidence/ccsdt/orca_crosscheck.py`, `evidence/ccsdt/README.md`, `evidence/ccsdt/MANIFEST.sha256`, `nlcsplit/tests/test_cc_convention.py`, `evidence/ccsdt/20261003T0703Z_stepm_water_fivebases_ghostcp_local.out`	the coupled-cluster anchor for the water-dimer binding energy: same official S22 geometry, same basis, same counterpoise convention, RHF reference, MP2/CCSD/CCSD(T) from PySCF's own modules, each rung reported both frozen-core and all-electron. The archive also records the mixed-convention bug this run nearly shipped with: freezing the monomers while switching only the dimer's convention printed -23.16 kcal/mol for the water dimer, which is impossible, and the impossibility was 17 kcal of oxygen 1s correlation counted on one side of the difference only. Two runs are kept because the second one exists to prove the first is still valid: after the frozen-core selection was refactored into a single call feeding both sides of the difference, the ladder was re-run with the script as it now ships, and stripped of wall-clock columns the two files differ in five lines, all of them one print statement whose wording changed - no energy moved. `nlcsplit/tests/test_cc_convention.py` pins that invariant, and one of its seven tests rebuilds the broken pairing on purpose to show the guard can fail. `evidence/ccsdt/orca_crosscheck.py` is the third-program check: ORCA 6.0.1, same geometries, same basis, same convention, on three systems and both conventions (eight rows, 24 single points), worst difference 0.0002 kcal/mol on the interaction energy and 1.2e-07 Ha between executions on the absolute energies. **What that check validated has to be restated:** it built its monomers the way the script of the day built them, so both programs were computing the *uncorrected* difference and agreeing about it - a cross-program check on a convention proves nothing about the convention when both sides share the same omission, which is why the corrected check is constructed from the definition instead of mirrored from the other script. The corrected comparison, with ghost functions on both sides and the basis dimension read back per part, is [S30]. `evidence/ccsdt/20261001T052943Z_orca_crosscheck_three_systems.out` is the one to cite for it; it also carries the correction of an earlier claim in this directory, which is why the basis check is stated the way it is. The first version of that check read ORCA's `Number of basis functions` field against PySCF's orbital count and reported agreement at 18 for a water monomer in 6-31G* - true, but not a criterion: for the same monomer in aug-cc-pVDZ the field prints 47 while ORCA's own SCF and coupled-cluster sections print `Basis Dimension Dim ....  41` and `Number of AO's ...  41`, which is PySCF's number. So the script now reads back the AO dimension per part and compares it with the PySCF nao for the same geometry and basis (24/24 equal), and its verdict refuses to print "agree" unless the energy test and the basis test hold at the same time. The 47 itself is

Tag	Artefact	What it fixes
S30	nlcsplit/step_m_ccsdt.py, nlcsplit/tests/test_cc_convention.py, evidence/ccsdt/README.md, evidence/ccsdt/20261002T1230Z_cp_defect_diagnosis.out, evidence/ccsdt/snnu_cp_defect_diagnosis.py, evidence/ccsdt/20261003T0616Z_orca_ghost_probe.out, evidence/ccsdt/orca_ghost_probe.py, evidence/ccsdt/20261003T0620Z_orca_vs_pyscf_ghost_cp.out, evidence/ccsdt/orca_ghost_cc_probe.py, evidence/ccsdt/20261003T0605Z_stepm_dimers_ghostcp_local.out, evidence/ccsdt/20261003T0652Z_snnu_cp_rebuild_631.out, evidence/ccsdt/snnu_cp_rebuild_631.sh, evidence/ccsdt/20261003T0700Z_snnu_cp_rebuild_compare.out, evidence/ccsdt/cp_rebuild_compare.py, evidence/ccsdt/ladder_parse.py, evidence/ccsdt/ladder_parse_selftest.py, evidence/ccsdt/20261003T0653Z_ghost_monomer_scf_check.out, evidence/ccsdt/ghost_monomer_scf_check.py, evidence/ccsdt/MANIFEST.sha256	the counterpoise correction itself, and the evidence chain that holds it up. 20261002T1230Z is the 10-second diagnosis that found it (a water monomer at 18 basis functions against the complex's 36, -6.8559 against -4.8789). 20261003T0616Z decides ORCA's ghost-atom spelling by the basis dimension rather than by the manual, and keeps two rejected spellings plus the dropped-partner case as negative controls, because without them "the colon form gives Dim=36" would not be evidence of anything. 20261003T0620Z computes the corrected anchor in two programs and reports 0.0001 kcal/mol on the counterpoise difference and on the uncorrected one at both conventions, with the ghost construction carried into the correlation space (Dim 36 on every part) and the frozen-core field read back as `chemical core (2 el)` on a monomer whose second oxygen has basis functions but no charge. 20261003T0652Z is the corrected ladder for all eight complexes at 6-31G*, Slurm job 51023 (COMPLETED, ExitCode 0:0, 42 min 30 s, 16 CPUs on cpu1), whose two pre-run guards compare the *complex's* RHF reference - the side the fix does not touch - against the archived values (4.31e-11 and 1.14e-11 Ha). evidence/ccsdt/cp_rebuild_compare.py then reads the old and new archives side by side and refuses to print a table unless every complex CCSD(T) energy is identical to 1e-7 Ha: 22 of 22 comparable rows agree to 0.0e+00, so the whole change in the dE_CP column is attributable to the monomer construction and not to the machine, the thread count or the geometry parse. The same script emits the eight deviations quoted in §3.2 and the two-grid-level table, reading both sides from archives rather than from this row. 20261003T0653Z closes the remaining way a ghost monomer could be wrong without being noticed: every ghost and bare SCF converged, and every ghost energy lies *below* its bare counterpart, which is the variational direction a counterpoise construction must have - a check independent of the convergence flag itself. evidence/ccsdt/ladder_parse.py is the single parser both consumers use, and evidence/ccsdt/ladder_parse_selftest.py is its eight-case self-test; four of those cases are fed deliberately broken rows, and the one that matters here is a positive interaction energy, which an earlier version of the parser dropped silently because its regex spelled the sign as -?

Tag	Artefact	What it fixes
S31	evidence/ccsdt/20261003T1212Z_snnu_sandwich_avdz_ghostcp.out, evidence/ccsdt/20261004T0135Z_cp_rebuild_compare_derived7.out, evidence/l2cp/20261004T0120Z_snnu51025_dft_avdz_ASSERT.out, evidence/l2cp/20261003T1747Z_snnu51052-dft_cp_avdz_pi.out, evidence/l2cp/20261003T1747Z_snnu51052-sbatch.out, evidence/l2cp/20261004T0328Z_snnu51083-dft_cp_avdz_small.out, evidence/l2cp/20261004T0328Z_snnu51083-sbatch.out, evidence/l2cp/20261004T0903Z_wsl2121-small_crosscheck.out, evidence/l2cp/20261003T1721Z_snnu51050_dft_avdz_small_interleaved.out, evidence/l2cp/cp_version_runner.py, evidence/l2cp/20261003T0907Z_wsl2121-cp_levels.out, evidence/l2cp/20261003T0911Z_wsl2140-cp_levels.out, evidence/l2cp/20261003T0804Z_snnu51028-guard_FAIL.out, evidence/l2cp/cp_version_probe.py, evidence/l2cp/20261003T0928Z_cp_version_probe.out, evidence/l2cp/dtot_2121_aromatic.py, evidence/l2cp/20261003T1025Z_wsl2121-dtot_aromatic.out, evidence/ccsdt/20261003T0925Z_cp_rebuild_compare_2121.out, evidence/ccsdt/20261004T0110Z_cp_rebuild_compare_derived.out, evidence/l2cp/20261003T1025Z_wsl2121-dtot_aromatic.out, evidence/l2cp/dtot_2121_aromatic.py	the counterpoise column of the functional side, and the fact that it was two columns. 20261003T0804Z is the guard that found it: a cluster job that must reproduce a level-2 counterpoise value before spending time on aug-cc-pVDZ got -6.1526 where it was told to expect -6.1290 and stopped with rc=1, which is the whole point of a pre-spend guard. evidence/l2cp/cp_version_runner.py is then run twice on one machine - same file, same geometry, same basis, same OMP thread count, same numpy and python, only PYTHONPATH differentiating PySCF 2.12.1 from 2.14.0 - and its two outputs are 20261003T0907Z and 20261003T0911Z. The 2.14.0 run reproduces the 2026-09-28 batch (-6.129446 against the archived -6.129, -3.675086 against -3.6750, -0.465680 against -0.4660), so nothing in that batch was miscalculated: the paper had put two implementations in one column. evidence/l2cp/cp_version_probe.py reads both logs and refuses to print unless (i) every complex and bare-monomer grid is point-for-point identical between the versions, (ii) the production counterpoise and the script's own re-derivation of the ghost monomer agree to 1e-6, and (iii) all six (system, level) cells are present in both; its verdict is that the ghost atom grids are the only thing that differs, in six of six cells, and that the uncorrected interaction energies agree to 1e-6 kcal/mol. That last point was, at first, the only thing licensing 3.1's BSSE list, three of whose cells subtracted a 2.12.1 counterpoise value from a 2.14.0 uncorrected one. Borrowing that justification was not good enough, so the uncorrected side of the five aromatic cells is now measured in the same release (evidence/l2cp/dtot_2121_aromatic.py -> evidence/l2cp/20261003T1025Z_wsl2121-dtot_aromatic.out: -4.340866, -2.928830, -1.780045, -3.578184 and -3.185938 kcal/mol), which reproduces the older batch to the three decimals it prints and makes 3.1's list eight same-release numbers rather than six mixed ones. 20261003T0925Z is the same comparison's output for the eight-row deviation table of 3.2: the table is now generated by deviation_table() rather than carried across by hand, the identity check has grown from 22 to 28 rows (worst complex-energy difference 2.0e-10 Ha across two days and two Slurm jobs), and the grid-level table has grown from five systems to eight. 20261003T1212Z closes the last counterpoise anchor of the correlated side (the sandwich at aug-cc-pVDZ, both frozen-core conventions, Slurm job 51024 finished 12:12:57Z), which is what lets 3.2 quote a span and a ratio for that complex; 20261004T0135Z is the regenerated derived table for seven systems. 20261004T0120Z is a failure kept on purpose: the functional-side aug-cc-pVDZ job

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S32	`evidence/nlc_cp/nlc_bsse_measure.py`, `evidence/nlc_cp/20261006T0657Z_nlc_bsse_three_small_dimers.out`, `nlcsplit/step_h_s22.py`, `nlcsplit/logs/step_h.json`, `evidence/nlc_cp/MANIFEST.sha256`	the basis-set superposition error of dE_{NLC} itself, which §4.1 used to exempt by assertion instead of measuring it. The script evaluates the same nonlocal term twice on one frozen superposed density - monomers in their own basis, then monomers carrying the partner's functions with no charge (the ghost water monomer is printed at N=13832 grid points against 9192 for the bare one, which is what a ghost basis costs here) - and reports the difference at level 1, 6-31G*, for water, ammonia and methane: -0.0074, -0.0031 and +0.0075 kcal/mol. Its denominator is not chosen at readout: the `span` column is read from the production sidecar `nlcsplit/logs/step_h.json`, the same four-scheme attribution span §2.3 publishes for that cell (0.0435, 0.0290, 0.0040 kcal/mol), so the ratios 0.17, 0.11 and 1.87 are two archived numbers divided, not a new estimate. Before any of it ran, the no-ghost side had to reproduce the archived dnlc values it claims to be the same quantity as, at a 0.001 kcal/mol gate: 8.660e-15, 6.443e-05 and 3.898e-04 kcal/mol. Three cells are the whole coverage of this row - the three aromatic complexes and both benzene dimers carry no measured dE_{NLC} superposition error, and this row names that gap instead of leaving the reader to find it

Tag	Artefact	What it fixes
S33	evidence/common_grid/ownership_vs_quadrature.py, evidence/common_grid/20261006T085207Z_ownership_vs_quadrature_l1_small.out, evidence/common_grid/20261006T090714Z_ownership_vs_quadrature_l1_benzenex.out, evidence/common_grid/20261009T083325Z_ownership_vs_quadrature_l1_benzene_dimers.out, evidence/common_grid/20261009T083325Z_ownership_vs_quadrature_l1_benzene_dimers.out.rc, evidence/common_grid/NOTE-clock-skew.md, nlcsplit/partition.py, nlcsplit/nlc.py, nlcsplit/tests/test_nlc_closure.py, evidence/common_grid/MANIFEST.sha256	the measurement that separates the two things a "partition scheme" changes at once. One SCF, one grid and one set of weights are held fixed while only the ownership factors vary: the package's own multi-centre factor is evaluated on the common grid's coordinates, normalised point-by-point to sum to one over atoms, and handed to the decomposition function through its soft= path - so the quadrature cannot move and the reported spread is attribution alone. The quadrature side is then read separately, as the totals each scheme produces on its own grid with the same density matrix. Four guards are in the file: every point's soft weights must sum to 1 within 1e-8 or the run exits; the hard nearest-atom assignment on the common grid must reproduce the published census control value (water, -7.8950 kcal/mol, gate 0.005 - the archive prints a difference of 0.0000); the closure check is on throughout, so each block sum is compared against one monolithic evaluation over the same kept points rather than against itself; and the soft= path itself is pinned by nlcsplit/tests/test_nlc_closure.py, whose negative controls regroup the blocks so that one cross term is double-counted and then dropped, and require the guard to fire. The same runs print the discrete residual of the local term, beta times (sum W rho minus the electron number), which is the quantity the abstract's cancellation statement is about: -0.0013 to +0.0011 kcal/mol over eight cells at grid level 1. The verdict the file pre-registered before it was run was a two-way one - attribution comparable to or larger than quadrature keeps the name "attribution uncertainty", attribution near zero would have taken that name away - and all eight cells came out on the first side, with the printed ratios spanning 1.29 (methane) to 1046.15 (benzene...water). Two provenance details are kept in the archive rather than smoothed over. The two 24-atom cells ran on a cluster node under a Slurm job, so the header line that records the repository commit is **empty** - the uploaded tree carries no git metadata - and the commit that produced them is recorded outside the file instead, together with the byte-identical hashes of the three modules the run prints for itself (nlcsplit/partition.py, nlcsplit/nlc.py, nlcsplit/step_h_s22.py), which match the ones now shipped and the ones that produced the six smaller cells; what makes those eight cells one comparison is those hashes, not the commit string. And the run's own exit code is kept in a separate rc file because a shared marker got overwritten by a later kill once before. Two things this archive is not. Its water cell reads a pure-attribution span of 0.0433 kcal/mol against the 0.0435 published for the same cell in Table 3, and the difference is the grid, not the density - one number is measured on the common grid with

Tag	Artefact	What it fixes
S34	evidence/common_grid/grid_level_convergence.py, evidence/common_grid/grid_level_convergence_negative_control.py, evidence/common_grid/run_grid_convergence_archive.sh, evidence/common_grid/20261023T0527Z_grid_convergence_full_set.out, evidence/common_grid/20261023T0527Z_grid_convergence_full_set.out.rc, evidence/common_grid/20261023T0527Z_grid_convergence_scoped.out, evidence/common_grid/20261023T0527Z_grid_convergence_scoped.out.rc, evidence/common_grid/20261023T0527Z_grid_convergence_negctl.out, evidence/common_grid/20261023T0527Z_grid_convergence_negctl.out.rc, evidence/common_grid/20261021T091653Z_hirshfeld_crossfamily_l2_water.out, evidence/common_grid/20261021T094326Z_hirshfeld_crossfamily_l2_benzwater.out, evidence/common_grid/20261021T140445Z_hirshfeld_crossfamily_l2_three_cells.out, evidence/common_grid/20261022T085504Z_hirshfeld_crossfamily_l2_three_pi_cells.out, evidence/common_grid/20261021T092258Z_hirshfeld_crossfamily_l0_currentcode.out, evidence/common_grid/20261021T094319Z_hirshfeld_crossfamily_l0_pisystem_currentcode.out, evidence/common_grid/20261012T050216Z_hirshfeld_crossfamily_l1_benzwater_after_unitfix.out, evidence/common_grid/hirshfeld_crossfamily.py, nlcsplit/partition.py, evidence/common_grid/MANIFEST.sha256	the grid-convergence ladder of the attribution spread, read-only and recomputed from the archives rather than transcribed. Every archive the ladder admits, excludes or splits is named inside the reading file itself, one line each with its byte count and digest, so this column names only the ASCII-spelled ones - seven of the eight level-1 rungs carry the system name in Chinese in their own filenames and pdflatex cannot set them; the reader reaches them from the reading file rather than from here. The gate is 0.005 kcal/mol, the same value the census guardrail prints inside these archives, and it is a constant in the script, not a function of the grid level being tested. Nothing here is trusted on its notification: an archive is admitted only if its own sidecar exit-code file, an in-file exit line, and a success code all agree, which is why 18 archives glob into the directory, 15 are admitted and 3 are excluded and named with the reason (one predates the grid-coordinate fix, two have a sidecar but no in-file exit line). Rungs are grouped by partition-file digest prefix and system, so a coarse reading from one revision of the code can never be joined to a fine reading from another and the pair that produced Table 3's two benzene rows under the pre-merge hash shows up as single-rung, i.e. not measured. Repeated runs at the *same* level are printed as a repeatability line and are explicitly not a convergence rung. Every span the table recomputes is tied out against the span printed in the same file, allowing one unit of the last printed digit: 52 comparisons, maximum deviation 1, none over tolerance - the residual is rounding, and the prose quotes the printed value. Two claims are reported and cannot be quoted for each other. The registered primary one asks whether each ladder is stable over its whole span of levels and answers **rc=3**, because two ladders exceed the gate on their coarse step (benzene...water 0.0076, benzene dimer (A) 0.0154 from level 0 to 1); that exit code is not softened, and level 0 is not dropped from the ladder to make the table pass. The secondary claim, added after the coarse steps were seen and printed as such, restricts the same gate to the top two rungs and only when both are at or above the published level - a level established by census of the certificate headers in the same directory, every one of which declares the same one, rather than a number typed by whoever wrote the script - and of the eight cells five have such a pair and all five are stable (0.0000, 0.0000, 0.0001, 0.0009, 0.0001); the other three are counted as NOT MEASURED rather than as passes, because their best pair reaches down to level 0 and therefore cannot answer the question. evidence/common_grid/grid_level_convergence_negative_control.py is the must-fail half of this row: ten cases, each with the exit code the script is required to

Tag	Artefact	What it fixes
S25	`nlcsplit/fastpair.py`, `nlcsplit/step_j_scaling.py`, `evidence/scaling/step_j.json`, `evidence/scaling/20260929T1340Z_step_j_scaling.out`, `evidence/scaling/step_j_rerun_20260930.json`, `evidence/scaling/step_j_rerun_20260930.txt`, `evidence/fastpair_ab/fastpair_ab_compare.json`, `evidence/fastpair_ab/fastpair_ide_validation.json`, `nlcsplit/tests/test_fastpair.py`, `evidence/scaling/MANIFEST.sha256`	the wall-clock and certificate numbers in §3.6, recomputed from these sidecars rather than transcribed into the prose: 33 screened/exact pairs over two windows, 1.08-10.28x (archived, 18 pairs) and 1.37-13.29x (re-run, 15 pairs), none faster than exact; the minimum block-pair certificate 197.96 Ha against the -8.34e-4 Ha quantity it would decide on, with n_pairs_bound_below_value = 0 and dropped_pairs = []. Two windows are kept because the second was measured on a different day and the two must not be merged into one interval. Each row carries two different error quantities and conflating them changes the headline: cert_err_ha is the guarantee the traversal can actually use, abs_dE_kcal is the measured deviation from the exact sum. Restricted to tau = 1e-12, exactly 8 of the 11 rows have abs_dE_kcal <= 1e-6 kcal/mol and those span 84.4-95.4% of far-field point pairs evaluated; only 4 rows have a *certificate* that small, and those span 93.7-95.4%. The 84.4% figure is therefore the usable-accuracy statement, not a mistake, and the looser certificate is precisely the result being reported. The monotone accept rule and the box separation are pinned by nlcsplit/tests/test_fastpair.py, which fails if a leaf pair is accepted out of order

Tag	Artefact	What it fixes
S27	`evidence/ccsdt/20261002T0715Z_snnu_benzene_avdz.out`, `evidence/ccsdt/snnu_benzene_avdz.sh`, `evidence/ccsdt/20261002T043400Z_snnu_benzene_ladder.out`, `evidence/ccsdt/20261009T085405Z_cp_rebuild_compare_frozengap.out`, `evidence/ccsdt/MANIFEST.sha256`	the second basis for the three π complexes, i.e. the three spans quoted in §3.2, Slurm job 50946 (COMPLETED, ExitCode 0:0, 44 min 22 s, 16 CPUs on cpu1), same script and same call path as [S24] and [S26], with a first-stage guard that re-derives the archived water-dimer aug-cc-pVDZ value (-5.2478 against -5.248). Values are read from the archived detail blocks rather than transcribed, and which construction a value came from has to be said out loud now: the six numbers this job printed for the three π complexes - frozen-core -4.5172 / -3.7273 / -3.0093 and all-electron -4.7192 / -3.9240 / -3.2010 kcal/mol, in benzene-water / benzene-ammonia / benzene-methane order - are monomers built in their own basis, the construction [S30] retired, and against the uncorrected 6-31G* rows of [S26] they give the 0.864 / 1.454 / 1.951 this row used to carry. Under ghost functions the same six cells are -2.7107 / -1.8795 / -1.1200 and -2.7144 / -1.8824 / -1.1225, so the three spans that enter the §3.2 comparison are 0.7383 / 0.9913 / 1.2016 kcal/mol against the corrected 6-31G* rows, every one of them printed by `evidence/ccsdt/20261009T085405Z_cp_rebuild_compare_frozengap.out` rather than subtracted by hand. That run retells the frozen-core story too, and the retelling is a correction to this row: the gap between the frozen-core and all-electron counterpoise value of one complex at one basis was 0.0354-0.0427 kcal/mol at 6-31G* and 0.1917-0.2020 at aug-cc-pVDZ in the own-basis column, while the ghost column puts those same six gaps at 0.0026-0.0032 and 0.0025-0.0037. An earlier version of this row reported that basis dependence as a property of the frozen-core choice. It is not one - it is the correlated BSSE changing size with the basis, the same effect §3.2 measures on the deviation, and in the corrected column no basis trend is left to read. The runner script is a byte- and hash-verified copy of the remote producer. No CBS claim: two basis points, no extrapolation

Tag	Artefact	What it fixes
S28	`evidence/ccsdt/20261002T0630Z_snnu_benzene_dimer_ccsdt.out`, `evidence/ccsdt/snnu_benzene_dimer.sh`, `evidence/ccsdt/20261002T0905Z_snnu_benzene_dimer_dft_cp.out`, `evidence/ccsdt/snnu_dft_cp_dimer.py`, `evidence/ccsdt/MANIFEST.sha256`	the parallel-displaced benzene dimer (S22_11, 192 AO) as the seventh anchored complex: the correlated side and the functional side under one basis and one convention. Both archives named in this row, and the runner script beside them, call that geometry "sandwich"; the word predates the measurement now kept in `evidence/recompute/20261005T0340Z_benzene_dimer_label_measure.out` (interplanar angle 0.0 deg, in-plane slip 1.698 A, which is what parallel-displaced means), and the archived bytes stay as they are because their hashes are registered - the dated note recording the correction is `evidence/ccsdt/README.md`. CCSD(T) is Slurm job 50945 (COMPLETED, ExitCode 0:0, 11 min 48 s, 32 CPUs), the DFT counterpoise values are job 50969 (COMPLETED, ExitCode 0:0, 38 min 07 s, 16 CPUs), and both runs first re-derived an archived value before spending the rest of their wall-clock - 50945 reproduced the water-dimer anchor to 0.0001 and the benzene-methane one to 0.0000 kcal/mol, 50969 reproduced benzene-methane at level 2 to 0.0000. Frozen-core counterpoise as those two jobs printed it: CCSD(T) -1.4999, ωB97X-V/6-31G* level 2 -1.9403 (deviation -0.440) and level 1 -0.7233 (deviation +0.777). Those two deviations are superseded rather than dropped: that CC row came from monomers solved in their own basis, and under the ghost-function construction of [S30] the same complex energy gives +1.4712, so the deviations become -3.4115 and -2.1945, which is what the table in section 3.2 carries. The same caveat applies to the rung-by-rung sentence after it: the corrected rebuild re-ran the CCSD(T) rung on this complex but did not print MP2 or CCSD for it, so those two rungs are on record only under the old monomer construction. The ladders also carry a result about correlation order that the deviation column cannot show: CCSD alone gives -0.6831 and the perturbative triples supply 0.817 of the -1.4999 total, while MP2 overbinds by a factor of two at -3.1059, so the two cheaper rungs err in opposite directions on the same system. Neither archive claims a CBS limit, a relaxed geometry, or that level 2 is grid-converged. This complex did get its second basis afterwards (`evidence/ccsdt/20261003T1212Z_snnu_sandwich_avdz_ghostcp.out`, named in [S31]), so its basis move and its ratio of 1.02 are quoted in section 3.2; what still has no correlated anchor of any kind is the third geometry, the one built by the benzene_sandwich() constructor of `nlcsplit/geomlib.py`, and Figure 5's lower panel stays at the six systems it was drawn with

Tag	Artefact	What it fixes
S29	`evidence/ccsdt/20261002T1030Z_snnu_benzene_dimer_B.out`, `evidence/ccsdt/snnu_benzene_B.sh`, `evidence/ccsdt/snnu_dft_cp_dimer_B.py`, `evidence/ccsdt/MANIFEST.sha256`	the eighth and last complex, the T-shaped benzene dimer (S22_20, 192 AO), which closes the counterpoise coverage of §3.2 at all eight systems - that archive's own header labels this geometry parallel-displaced, the same inverted pair of words the row above documents, and the measured interplanar angle of 90.0 deg is what makes it T-shaped. One Slurm job (50980, COMPLETED, ExitCode 0:0, 1 h 02 min 23 s, 16 CPUs) ran both sides: CCSD(T)/6-31G* frozen-core -2.5506 and all-electron -2.6300, and ωB97X-V/6-31G* counterpoise -2.4297 at level 2 (deviation +0.121) against -2.6953 at level 1 (deviation -0.145). Two guards ran first and are quoted in the archive: the water-dimer anchor to 0.0001 kcal/mol and the sandwich's own CCSD(T) value to 0.0000, the latter being another cross-job reproduction of `evidence/ccsdt/20261002T0630Z_snnu_benzene_dimer_ccsdt.out`. The grid step on this complex is 0.266 kcal/mol and changes the sign of the deviation, which is why the grid paragraph counts three sign changes out of the five systems measured at both levels. No CBS limit, no relaxed geometry, and no claim that level 2 is grid-converged: the archive states all three as non-claims

Tag	Artefact	What it fixes
S26	`evidence/ccsdt/20261002T043400Z_snnu_benzene_ladder.out`, `evidence/ccsdt/snnu_benzene_runner.sh`, `evidence/ccsdt/20261002T0528Z_snnu_benzene_dft_cp.out`, `evidence/ccsdt/snnu_dft_cp_levels.py`, `evidence/ccsdt/20261002T0620Z_snnu_benzene_methane_l2.out`, `evidence/ccsdt/snnu_dft_cp_methane_l2.py`, `evidence/ccsdt/MANIFEST.sha256`	the benzene-complex ladder behind §3.2's third and fourth paragraphs, computed on a second cluster after the first one saturated. The CCSD(T) side is `nlcsplit/step_m_ccsdt.py` unchanged, run under Slurm on partition `cpu` (job 50937, COMPLETED, ExitCode 0, 4 min 08 s) through the same call path as the water/ammonia/methane ladder of [S24], which is what makes the two comparable rather than merely similar; the producer script is archived verbatim beside the output. Before any benzene wall-clock was spent, the runner re-derives the already-archived water-dimer value from the same code and refuses to proceed otherwise (`guard: |-6.8559 - (-6.856)| = 0.0001 <= 0.0015 -> PASS`), and the archive records the two ways that guard first failed in ways that both looked like success: a 0-byte uploaded script that the scheduler reported as COMPLETED with exit 0 in one second, and a guard that parsed a summary table the script does not print for a single system. The DFT side deliberately imports the counterpoise routine of `nlcsplit/step_h_s22.py` instead of running that module, because its output path is hardcoded to `nlcsplit/logs/step_h.json`, the carrier of the main table - three numbers are not worth overwriting a published archive. The level-2 grid claim rests on three archives, not two: the first DFT run (job 50938) was cancelled by a 30-minute limit and declares itself partial in its own header, and the missing benzene-methane row came from job 50943, whose archive also records that its level-1 line reproduces the earlier job byte for byte. The producer scripts were copied only after their sha256 matched the remote originals. None of the three archives claims a CBS limit (one basis, no extrapolation) or relaxed geometries (official S22 initial structures, unrelaxed), and the benzene dimer at 192 AO is in none of them
S35	`evidence/chemical_controls/dative/run_ab.py`, `evidence/chemical_controls/dative/ab_nlc.json`, `evidence/chemical_controls/dative/ab_opt.xyz`, `evidence/chemical_controls/dative/ab_opt.out`, `evidence/chemical_controls/dative/ab_complex.out`, `evidence/chemical_controls/dative/ab_nh3.out`, `evidence/chemical_controls/dative/ab_bh3.out`, `evidence/chemical_controls/dative/ab_run.log`	the dative H3N-BH3 control: B-N 1.6677 A, four-scheme span 0.0710 kcal/mol at levels 1 and 2, and the ORCA 6.0.1 dE_{NLC} -1.9711 kcal/mol against the PySCF -1.9748/-1.9753 values

Tag	Artefact	What it fixes
S36	evidence/chemical_controls/ionic/run_fw.py, evidence/chemical_controls/ionic/fw_nlc.json, evidence/chemical_controls/ionic/fw_opt.xyz, evidence/chemical_controls/ionic/fw_opt.out, evidence/chemical_controls/ionic/fw_complex.out, evidence/chemical_controls/ionic/fw_f.out, evidence/chemical_controls/ionic/fw_water.out, evidence/chemical_controls/ionic/fw_run.log	the ionic F-...H2O control: F-O 2.4604 A, four-scheme spans 0.0828/0.0825 kcal/mol at levels 1/2, and the ORCA 6.0.1 dE_{NLC} -0.9850 kcal/mol against the PySCF -0.9875/-0.9877 values
S37	evidence/release_qa/20261003T1901Z_quickstart_readme_example.out, evidence/release_qa/20261013T150059Z_shipping_wheel_census.out, evidence/release_qa/20261013T150059Z_shipping_wheel_census.out.rc, evidence/release_qa/20261013T153804Z_test_count_census.out, evidence/release_qa/20260929T213552Z_no_gamess_scan_after_allow.out, evidence/release_qa/MANIFEST.sha256, nlcsplit/examples/quickstart.py, tools/make_public_repo.js, CITATION.cff	the release record of the package, which is what the appendix "The nlcsplit package" is written against: the quickstart worked example run verbatim, with the interpreter version and the SHA-256 of the four modules that produced it; the wheel census, which prints the member categories beside their sum and records that the nlcsplit/partition.py inside the wheel is byte-identical to the one in the working tree; the test-count census, which sums to its own total and names the modules that stay out of the wheel instead of leaving them silent; the licence scan, which walks the files an install actually produced rather than grepping the source for a word, and lists its three named exemptions at exit 0; and the two generators those archives come from. The counts quoted in that appendix are the counts of these archives at the revision each header records; the census re-runs at every export, so a reader comparing a later wheel should expect the shipped list to have grown rather than to disagree. What this row does not support: anything about the peer's frozen reference implementation, which is named in the export tool's internal-only list and is not distributed, and anything about a tagged release or deposit DOI, which do not exist yet.

Three absences are stated rather than hidden. evidence/centos/...step_f_h2o2.out predates the water-dimer geometry fix (its inter-fragment minimum is 2.0258 A against 1.95 A in evidence/l2cp/), so S8 cites it only for the component accounting, never as a current-geometry result. S15's files were sitting in scratch/, which is gitignored - they were promoted into evidence/ on 2026-09-29 for exactly that reason. And one distinction has to be kept: S6 and S13 come from the same CentOS batch timestamped 2026-09-28T16:58, which is earlier than the geometry fix, while S12's primary artefact is a recompute run afterwards. Within that earlier batch the initial S22 structure and the relaxed one are distinguishable (the level-0 6-31G* value -0.5316 appears in S13's ladder, while the -0.5975 of the relaxed structure does not), so S13's cross-grid spreads are read here as measured on the *initial* S22 geometry, not on the relaxed one that carries the headline numbers. The level-1 column of that same ladder agrees with the later recompute to four decimals for all three bases the two runs share, so the grid-convergence conclusion does not turn on which geometry was used - but that was checked after the fact. The full ladder has since been re-run on the geometry that carries the headline numbers, and section 3.4 quotes that run (evidence/recompute/20260930T191014Z_stepg-official.out); what remains true is the weaker thing: the

cross-grid spreads of S13 are still measured on the initial structure, because that is the batch which contains all four grid levels.

Hirshfeld cross-check status. The full-S22 cross-family check is not used as evidence in this paper. Its Hirshfeld column agrees with the older eight-cell archive in four of the eight cells; the other four (#10, #11, #18 and #20) are reported as NOT MEASURED for equivalence rather than reconciled, and no number in Tables 1-3 or Table S2 depends on that column. Because the column is outside the four-scheme family, its absence cannot move the reported range.

Family-internal span over the full S22 set. The table below is the full S22 survey referenced in section 3.4; values are parsed from the archived certificates by `summarize_s22_full.py`. It uses only the four family-internal Becke-type schemes and is numbered Table S2 in the Supporting Information.

Table S2. Family-internal attribution span over all 22 S22 complexes at grid level 1, 6-31G*, for the four Becke-type schemes. Values are parsed from the archived certificates by `summarize_s22_full.py`; this table is numbered Table S2 in the Supporting Information.

S22	fragments	atoms	family-internal span (kcal/mol)
S22_01	H3N + H3N	8	0.0284
S22_02	H2O + H2O	6	0.0433
S22_03	CH2O2 + CH2O2	10	0.2342
S22_04	CH3NO + CH3NO	12	0.2143
S22_05	C4H4N2O2 + C4H4N2O2	24	0.2152
S22_06	C5H5NO + C5H6N2	25	0.2716
S22_07	C5H5N5 + C5H6N2O2	30	0.2664
S22_08	CH4 + CH4	10	0.0034
S22_09	C2H4 + C2H4	12	0.0292
S22_10	C6H6 + CH4	17	0.2142
S22_11	C6H6 + C6H6	24	0.0274
S22_12	C4H4N2 + C4H4N2	20	0.0098
S22_13	C4H4N2O2 + C4H4N2O2	24	0.0260
S22_14	C6H6 + C8H7N	28	0.0409
S22_15	C5H5N5 + C5H6N2O2	30	0.0381
S22_16	C2H4 + C2H2	10	0.0490
S22_17	C6H6 + H2O	15	0.2469
S22_18	C6H6 + H3N	16	0.2219
S22_19	C6H6 + CHN	15	0.3030
S22_20	C6H6 + C6H6	24	0.2196
S22_21	C6H6 + C8H7N	28	0.2556
S22_22	C6H6O + C6H6O	26	0.1407

The nlcsplit package

This appendix describes the code that produced the numbers above. It is filed here rather than as a separate software manuscript because everything method-shaped in such a manuscript would repeat section 2 and everything result-shaped would repeat section 3, and a second paper would then need a second provenance system to keep the two in step. Everything stated below is checkable from the repository: the entry point by importing it, the release record by reading the archives named in [S37].

Shape. `nlcsplit` is pure Python on NumPy and PySCF. No compiled extension is built or required, and those two are the only dependencies. Three modules carry the physics - `nlcsplit/nlc.py` (the VV10 fields and the blocked pair sum), `nlcsplit/partition.py` (the attribution schemes) and `nlcsplit/geomlib.py` (geometries, fragment boundaries derived from cova-

lent connectivity, and the loader for the published S22 geometries). Every table of the manuscript is produced by exactly one `step_*.py` script, which writes a JSON sidecar beside its archived standard output, and `nlcsplit/figures.py` draws only from those sidecars and refuses to draw when one fails to parse. A figure therefore cannot outlive the run that made it, which is the failure mode the whole archive structure exists to prevent.

One entry point. The public surface is a single function:

```
nlc.fragment_decomposition(coords, weights, rho, sigma, fragments, b, C,
owner=None, soft=None, rho_floor=1e-12, closure_check=False)
```

`fragments` is a list of atom-index groups. `owner` gives the index of the atom that owns each grid point - a hard assignment - and `soft` is an (natm, ngrid) weight matrix whose columns sum to one, as returned by `partition.scheme_factors`. The function returns `E_loc`, `E_nl`, `E_total`, `intra` keyed by fragment and `inter` keyed by fragment pair, and `inter[(A, B)]` is the whole cross contribution rather than a half, which is what makes the sum of `intra` plus the sum of `inter` equal `E_nl` in the first place. `closure_check=True` compares that block sum against one monolithic evaluation over the same kept points and raises if they differ; it is off by default because the reference doubles the summation, and section 2.2 states both the residual it gives on the systems reported here and the inputs the check shares with the quantity it audits [S33]. Parameters come from PySCF's `nlc_coeff` rather than from constants in this package, so a different VV10 parameterisation cannot silently disagree with the functional that generated the density.

Six attributions, one shape. `partition.scheme_factors(mol, coords, scheme)` dispatches on the four Becke-family constructions used in the text (`becke`, `becke-becke-rad`, `becke-treutler`, `stratmann`) and on two cross-family controls (`voronoi`, `hirshfeld`). All six return the same array shape, so comparing attributions is a one-keyword change and nothing else moves. Every construction passes the same guard before returning: the pointwise sum of the attribution weights must sit within 1e-12 of one, or the function raises instead of handing back a matrix that would silently mis-normalise the double sum. That figure is an assertion threshold in the shipped code, not a measured residual, and `nlcsplit/tests/test_partition_factors.py` is where it is pinned. The two cross-family constructions enter no span quoted in this paper: they are reported separately and are thread-count sensitive, which is recorded in the provenance row for the grid ladder [S34], and the ranges the text publishes are family-internal by construction.

The worked example, run verbatim. `nlcsplit/examples/quickstart.py` on the water dimer at omegaB97X-V/6-31G*, grid level 1, with the Becke partition, prints

```
inter[(0,1)] = -0.5237 intra = -3.5457 -3.7437 E_nl = -7.8130 E_loc = 60.8303
E_total = 53.0172
```

in kcal/mol, and the archive of that stdout carries the interpreter version and the SHA-256 of the four modules that produced it [S37]. Two arithmetic points are worth stating rather than skipping, because the printed lines invite the reader to add them: the three block values sum to one unit in the last printed place above the printed `E_nl`, and `E_nl` plus `E_loc` differs from the printed `E_total` by the same one unit. Each line rounds independently, and the additivity guard of section 2.2 runs on the doubles rather than on what is printed. `E_total` is a different object from the interaction energy of section 3 and must not be read as one; the agreement with the native kernel that the text quotes is the 5.7e-06 kcal/mol of section 2.1, measured on a fixed density and a fixed grid [S5].

Two checks that ship with the package. Neither is about this paper's conclusions; both are about whether the quantity is what the label says. The first recomputes the nonlocal term for the eight complexes in an external program, ORCA 6.0.1, together with an omegaB97X control that must not print the nonlocal label at all - a gate on the reporting convention rather than on a number [S18]. The second re-evaluates the cross term of a two-Gaussian model density by three independent routes and reports its own error bar, so the attribution can be audited on a

system where no real-space partition is involved at all [S22]. The two chemical controls of section 3.4 are ordinary calls of the same pipeline on a dative and an ionic complex [S35, S36], and no new attribution code was written for either.

What runs before a release. The package is Apache-2.0 and contains no GAMESS source. That claim is checked by scanning the files an install actually produces for source artefacts - `*.src`, `*.f`, `*.for`, `GAMESS*` - rather than by grepping the tree for a word the disclaimer itself contains. The archived scan reports 45 installed files, three named exemptions and no hit outside the exemption ledger at exit code 0 [S37]. Correctness is guarded in the same spirit: the test suite pins the kernel against PySCF's native one, regression-tests the archived numbers, and checks the geometry and partition-factor invariants; the suite collects 94 tests at the revision that census records (71 not marked slow, 23 marked slow), it prints its per-module counts beside their total so that a silently dropped category cannot pass as a member count, and it names the modules that stay out of the wheel with the reason rather than leaving them absent [S37]. Above that, the export tool `tools/make_public_repo.js` enforces fifteen gates over the tree that would actually be published, and one command in the parent repository (`tools/run_ship_qa.sh`) drives it end to end. Each gate carries a two-directional self-test that must fire on a planted fault and stay silent on valid input, and the build refuses to ship when a self-test does not fire - a gate that cannot be made to fail is not a gate. Table S3 lists them with the real defect that put each one there. These are the counts of the archived run named above; the census re-runs against whatever wheel is actually released, so a reader holding a newer build should expect the shipped list to have grown rather than to disagree.

Table S3. The fifteen release gates of the export tool `tools/make_public_repo.js`: what each checks, and the defect that put it there. A letter identifies a gate in that file. Rows marked "planted only" have fired in self-test but not yet on real input; they are listed with the rest because the two-directional self-test is what makes them load-bearing, not the history. This table is numbered Table S3 in the Supporting Information.

gate	what it checks	how it earned its place
R	walks the export tree for build and pytest residue left inside it - `__pycache__`, `.pyc`, `.egg-info`, a stray `setup.cfg` - which is how a compiled docstring can carry a repository path out even when every source file is clean	13 `.pyc` files that pytest had left inside the release directory; the old check iterated the list of copied names instead of walking the tree, so it never opened them
A	no GAMESS source artefacts, matched on source patterns rather than on the word "gamess"	a permanent false positive on the package's own disclaimer, which is why it matches patterns and not the word
B	walks the published tree and fails by default on credentials and internal-network strings - private-key blocks, password assignments, ssh targets, bare-IP users - and, through the ten named rules that stay out of the release, on working-machine paths, bare cluster addresses and account names	13 published files carried working-machine paths while the gate reported every one of its arrays empty, because its patterns covered artefact extensions but not path forms
C	every `evidence/*/MANIFEST.sha256` reproduces byte for byte	a Windows path making `sha256sum` report six mismatches that were not there
C2	the publishing scrub changed no digit in any number, and every file in a published evidence directory is registered in that directory's own source manifest	the second half fired on real input the day it was written: a comparison JSON produced by a step script landed in `evidence/agentb_sweep/` with no manifest entry and the build refused to ship
D	every path cited by the provenance appendix is actually shipped	three citations that resolved to nothing in the published tree

gate	what it checks	how it earned its place
E	every import in a shipped module is declared in `pyproject.toml`	`nlcsplit/figures.py` importing matplotlib, which no dependency listed
F	the wheel and sdist member lists are what the release claims they are	two script-style tests that survived a build already reported fixed
G	`LICENSE` in the source file carries the real copyright line, not the Apache placeholder	the legal statement existed only in the generated copy; fires today on a restored placeholder
T	nothing in the export tree is missing from the export manifest	added as a belt-and-braces check; no firing on real input is recorded in this repository, only on the faults planted in its self-test
U	every module a shipped file imports from `nlcsplit` - absolute, dotted or package-relative - is itself in the shipped tree, with a self-test against a planted missing sibling	written for `nlcsplit/fastpair.py`, which gate E could not see because an intra-package import is not a declared dependency
V	every Python module inside a packaged directory is either in the export whitelist or named, with a reason, as an unreleased internal module - the set that `pip install .` actually puts in `site-packages`, which the whitelist does not constrain	found by the licence scanner of gate A, which walked 45 installed files from a source-directory install and found the peer implementation among them
W	no manuscript or README quotes a share for a system whose denominator cancels, that is where the ratio of the two terms exceeds two in absolute value - the rule `nlcsplit/step_f_dispersion.py` already applies to its own output, extended to the prose	three of them on the day it was written, in the companion text's conclusion, this paper's section 4, and a README line the author had just added; the gate then caught its own author reintroducing the number while documenting the gate
X	every `step_*.py` in the export tree appears in the driver's command table, so each table in the text has an entry point a reader can invoke	`step_m_ccsdt.py` reached the release on 2026-09-30 without ever being registered, which is the same omission the driver had been rewritten to fix and that rewrite left no guard; the self-test truncates the table to the four entries of the original bug and must then report ten missing scripts
Y	every image and link target in the short-form software draft resolves to a file the export actually carries, walking the copied-file set rather than the filesystem	planted only: it has not fired on real input, its self-test must fire on exactly the planted target and stay silent on a valid one, and it goes quiet when that draft is absent from the tree

Limits of the software. The pair sum is quadratic in the number of retained grid points and there is no sub-quadratic approximation in the package. One attempt at a certified screening was built, measured and rejected; section 3.6 states the measurement and what it does not license, and the companion screening manuscript carries the raw timings. The implementation is VV10 only - the tau-dependent rVV10 form is not implemented, and the parameters are read from whatever the installed PySCF reports for the functional in use, so a functional whose VV10 parameters PySCF does not expose cannot be evaluated here. And this is not an energy decomposition analysis: it resolves one term of the functional into fragment pairs and says nothing about electrostatics, exchange-repulsion, polarisation or charge transfer, which is why section 4.1 states what the decomposition is not before section 4.2 lets the reader use it.